# SOC Dashboard Project Report

***Failed Login Monitoring & Security Analysis using Splunk***

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### 1. Project Overview

This project focuses on creating a **Security Operations Center (SOC) Dashboard using Splunk** to monitor and analyze failed login activities.

The dashboard helps SOC analysts identify:

- Total failed login attempts
- Users with repeated failed logins
- Failed login activity over time
- Source IP addresses generating failed logins

**Main Goal**

The primary goal is to identify suspicious authentication activities such as **brute-force attacks, password guessing, and abnormal login behavior**.

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### 2. Log Ingestion into Splunk

The security logs were first imported into Splunk and stored in the required index.

**Steps**

**Splunk → Add Data → Select Log Source → Configure Source Type → Select Index → Submit**

After ingestion, the logs were searched and verified to ensure that Splunk was receiving the events correctly.

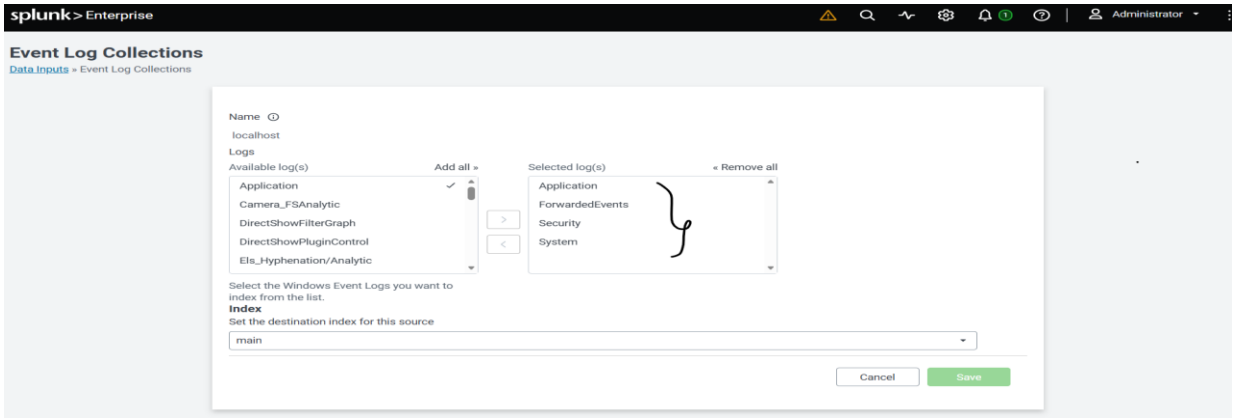



*Figure 1: Log ingestion in Splunk*

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**3. Log Field Identification**

After loading the logs, the important fields required for security analysis were identified.

| Field | Purpose |
|---|---|
| Timestamp | Identify when the event occurred |
| Username | Identify targeted users |
| Source Network Address | Identify the source IP |
| Status/Event | Identify failed login activity |

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**4. SOC Dashboard – Failed Login Monitoring**

The **Failed Login Monitoring Dashboard** contains four panels that provide different views of authentication failures.

Together, these panels help the SOC analyst understand **how many failures occurred, who was targeted, when the activity occurred, and where it originated.**

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**Panel 1 – Total Failed Login Attempts**

**Purpose**

Shows the **total number of failed login attempts** during the selected time period.

**SPL Query**

index=<your_index> "failed login"

| stats count as "Total Failed Attempts"

**Visualization**

**Single Value**

**Security Use**

Helps the SOC analyst quickly determine whether the environment is experiencing a **high volume of failed authentication activity**.

**Panel 2 – Top Users with Failed Logins**

**Purpose**

Shows the users who have the **highest number of failed login attempts**.

**SPL Query**

index=<your_index> "failed login"

| stats count as failed_attempts by user

| sort - failed_attempts

| head 10

**Visualization**

**Bar Chart**

**Security Use**

Helps identify accounts that may be targeted by **password guessing, brute-force attacks, or password spraying**.

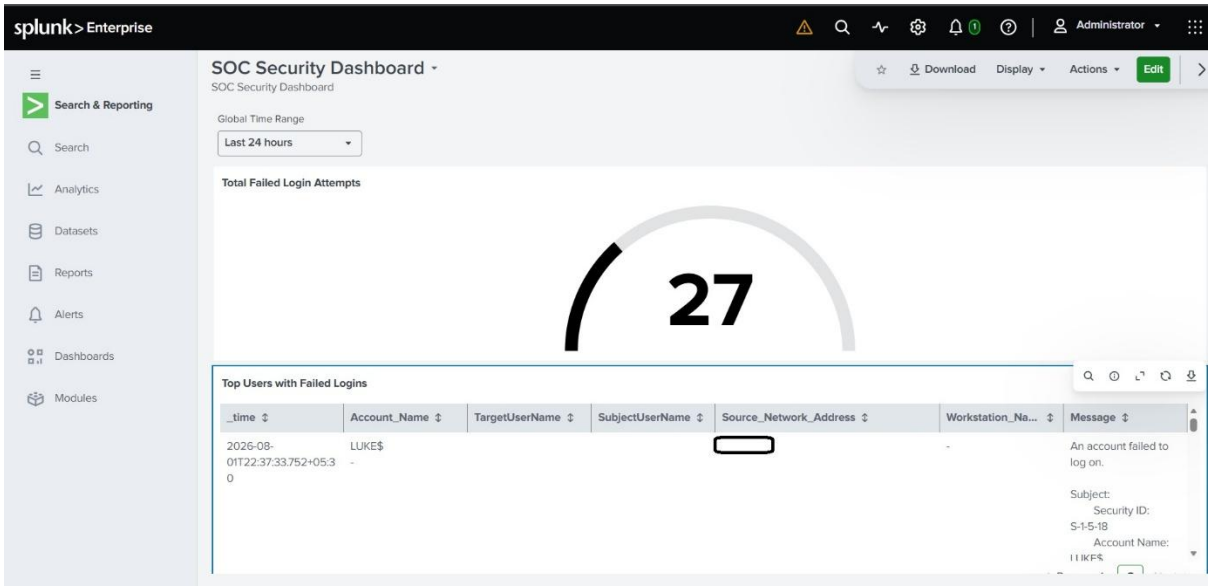



*Figure 2: Total failed login attempts and Top Users with Failed Login Attempts*

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**Panel 3 – Failed Login Trend**

**Purpose**

Shows failed login activity **over time**.

**SPL Query**

index=<your_index> "failed login"

| timechart count as failed_attempts

**Visualization**

**Line Chart**

**Security Use**

Helps identify **sudden spikes or unusual increases** in failed login activity.

A sudden spike may indicate automated or malicious authentication attempts.

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**Panel 4 – Top Source IP Addresses**

**Purpose**

Shows the **source network addresses** generating the highest number of failed login attempts.

**SPL Query**

index=<your_index> "failed login"

| stats count as failed_attempts by "Source Network Address"

| sort - failed_attempts| head 10

**Visualization**

**Bar Chart / Table**

**Security Use**

Helps identify suspicious sources that may be generating repeated authentication attempts.

This allows the analyst to answer:

**Where are these failed login attempts coming from?**

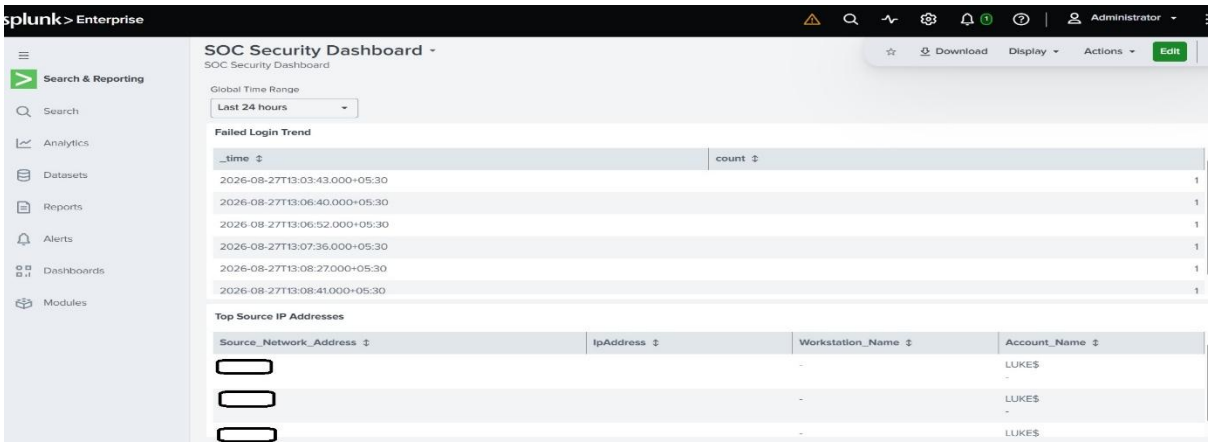



*Figure 6: Failed Login Trend and Top Source IP Addresses Panel*

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**5. Complete SOC Dashboard**

The four panels were combined into a single dashboard to provide a **centralized view of failed login activity**.




*Figure 4: Failed Login Monitoring SOC Dashboard*

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## 6. Dashboard Analysis

The dashboard allows the SOC analyst to move from **detection → analysis → investigation**.

### Investigation Flow

**High Failed Logins**
|
**Check Login Trend**
|
**Identify Targeted User**
|
**Check Source IP**
|
**Investigate Activity**

This workflow helps analysts quickly identify potentially suspicious authentication behavior.

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## 7. Security Alert

Alerts can be configured to automatically detect **excessive failed login attempts**.

### Alert Purpose

Detect unusually high failed login activity within the selected time period.

### SPL Query

index=<your_index> "failed login"

| stats count as failed_attempts

| where failed_attempts > 10

**Trigger Condition**

The alert is triggered when the number of failed login attempts **exceeds the defined threshold**.

**SOC Response**

When the alert is triggered, the analyst can investigate:

- Source Network Address
- Targeted User
- Number of Attempts
- Time of Activity
- Related Security Events

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**8. Conclusion**

The **Splunk SOC Dashboard** provides a centralized and effective way to monitor failed login activity.

It helps SOC analysts identify:

**High-volume failures + Targeted users + Suspicious trends + Source IPs**

This improves visibility into authentication activity and supports the **early detection and investigation of potential security threats**.